

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

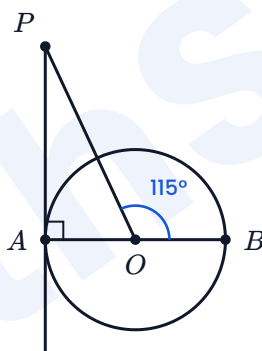
Chapter 10: Circles

Previous year board questions, topic-wise: 9 questions

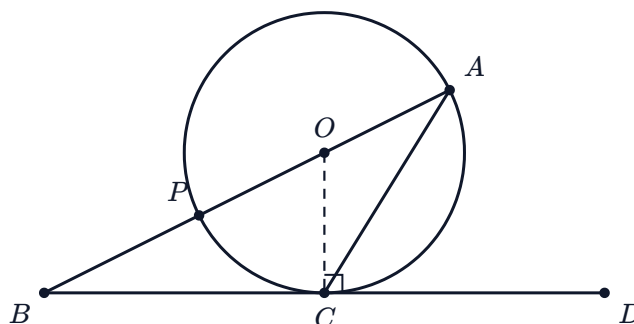
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Theorem 1: tangent and radius

1. Tangents are drawn to a circle at the two ends of a diameter. How are the two tangents related? **CBSE 2025, 1 mark**
2. AOB is a diameter of a circle with centre O , and PA is the tangent at A . If $\angle POB = 115^\circ$, find $\angle APO$. **CBSE 2025, 1 mark**



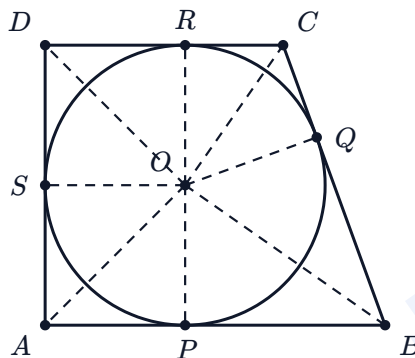
3. O is the centre of a circle and BCD is the tangent to it at C . The line from B through O meets the circle at P and A . Prove that $\angle BAC + \angle ACD = 90^\circ$. **CBSE 2025, 2 marks**



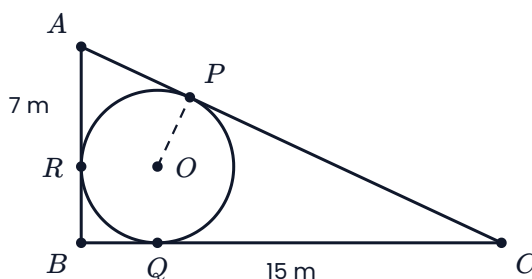
4. Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. **CBSE 2024, 1 mark**

2 Theorem 2: tangents from an external point

5. A point P is 26 m from the centre of a circle. Each tangent from P to the circle is 10 m long. Find the radius of the circle. **CBSE 2025, 2 marks**
6. A quadrilateral is drawn around a circle so that all four sides touch it. Prove that each pair of opposite sides subtends supplementary angles at the centre. **CBSE 2025, 2 marks**



7. AB is a diameter of a circle with centre O . The tangents at A and B are met by a third tangent at Q and P respectively. Prove that $\angle POQ = 90^\circ$. **CBSE 2024, 3 marks**
8. A circle of radius 8 cm with centre O touches the sides AD , DC , CB and BA of a quadrilateral $ABCD$ at P , Q , R and S . $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm. Find DC . **CBSE 2024, 3 marks**
9. A triangular backyard ABC has a right angle at B , with $AB = 7$ m and $BC = 15$ m. A circular pit inside it touches AC , BC and AB at P , Q and R , and $AP = x$ m. (i) Write AR in terms of x . (ii) What kind of quadrilateral is $BQOR$? (iii) Write PC in terms of x and find x , OR find x and then the radius r of the pit. **CBSE 2024, 4 marks, case study**



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